

Integration [Newton's Cotes Method]

Applying integration on Lagrange Formula:

$$I_n = \int_a^b P_n(x) = \int_a^b \underbrace{\sum_{k=0}^n l_k(x)}_{\text{weight function}} * f(x_k) dx.$$

$\downarrow$
degree

$\downarrow$
weight function, $\partial_k = \int_a^b l_k(x) dx$

if nodes are
equally distanced
 $\downarrow$
Newton's cotes
formula

So,

$$I_n = \int_a^b \sum_{k=0}^n l_k(x) * f(x_k) dx$$
$$= \sum_{k=0}^n \partial_k f(x_k) \leftarrow \text{formula}$$

closed Newton cotes formula:

$[a, b]$

$\downarrow$
both inclusive

$[1, 5]$

$\rightarrow 1, 2, 3, 4, 5$

$$h = \frac{b-a}{n}$$

$\downarrow$
h is step size!

Open Newton cotes formula

$[a, b]$

$\downarrow$
Not inclusive

$[1, 5]$

$[2, 4]$

$$h = \frac{b-a}{n+2}$$

Trapezium Rule / closed Newton's cotes formula ($n=1$)

As, degree $n=1$; So, there are two (2) nodes.

[Derivation]

$[a, b]$
 $x_0 \swarrow \searrow x_1$

$$P_1(x) = l_0(x)f(x_0) + l_1(x)f(x_1)$$

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - b}{a - b}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - a}{b - a}$$

We know

$$\delta_0 = \int_a^b l_0(x) dx = \frac{1}{a-b} \int_a^b (x-b) dx = \frac{1}{a-b} \left[\frac{1}{2}x^2 - bx \right]_a^b$$
$$\therefore \delta_0 = \frac{b-a}{2}$$

Similarly

$$\delta_1 = \int_a^b l_1(x) dx = \frac{1}{b-a} \int_a^b (x-a) dx = \frac{1}{b-a} \left[\frac{1}{2}x^2 - ax \right]_a^b$$
$$\therefore \delta_1 = \frac{b-a}{2}$$

Using δ_0, δ_1

$$\begin{aligned} I_n &= \int_a^b P_1(x) dx = \int_a^b l_0(x)f(x_0) + l_1(x)f(x_1) dx \\ &= \delta_0 f(x_0) + \delta_1 f(x_1) \\ &= \frac{b-a}{2} f(x_0) + \frac{b-a}{2} f(x_1) \end{aligned}$$

for $n=1$ $\rightarrow \therefore I_n = \frac{b-a}{2} (f(x_0) + f(x_1))$

Ex. Given, $f(x) = x^2 + 5x$; interval range $[0, 5]$

$x_0 = a$ $x_1 = b$

Ans.

* Solve using Trapezium Rule: [Numerical Integration]

$$\begin{aligned} I_n &= \frac{b-a}{2} (f(x_0) + f(x_1)) \\ &= \frac{5-0}{2} ((0^2 + 5 \cdot 0) + (5^2 + 5 \cdot 5)) \\ &= 125 \end{aligned}$$

* Actual Exact Integration will be :

$$\begin{aligned} &\int_0^5 x^2 + 5x \, dx \\ &= \left[\frac{x^3}{3} + \frac{5}{2} x^2 \right]_0^5 \\ &= \left(\frac{5^3}{3} + \frac{5}{2} \cdot 5^2 \right) - \left(\frac{0^3}{3} + \frac{5}{2} \cdot 0^2 \right) \\ &= 104.17 \end{aligned}$$

$$\begin{aligned} \text{Calculated Error} &= | \text{Actual} - \text{calculated value} | \\ &= | 104.17 - 125 | \\ &= 20.83 \end{aligned}$$

$I_n > \text{actual value}$ 
this is Overestimated result!

Ex. Calculating Error Bound

$$f(x) = e^x ; [0, 2] ; n = 1$$

Ans.

We know,

$$\text{Upper bound Error} = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} \right|_{\substack{\text{Max} \\ \xi \in [a,b]}} (x-x_0)(x-x_1) \dots$$

Now

$$\text{u.b. error} = \left| \frac{f^2(\xi)}{2!} \right|_{[0,2]} (x-x_0)(x-x_1)$$

$$= \left| \frac{1}{2!} e^2 \right| (x-0)(x-2)$$

$$= \left| \frac{1}{2} e^2 \right| \cdot \underbrace{x^2 - 2x}_{w(x)}$$

$$\int_0^2 w(x) dx = \int_0^2 x^2 - 2x dx = \left[\frac{1}{3} x^3 - \frac{2}{2} x^2 \right]_0^2 = \frac{4}{3}$$

plugging values

$$\text{u.b. error} = \frac{1}{2} e^2 \cdot \frac{4}{3}$$